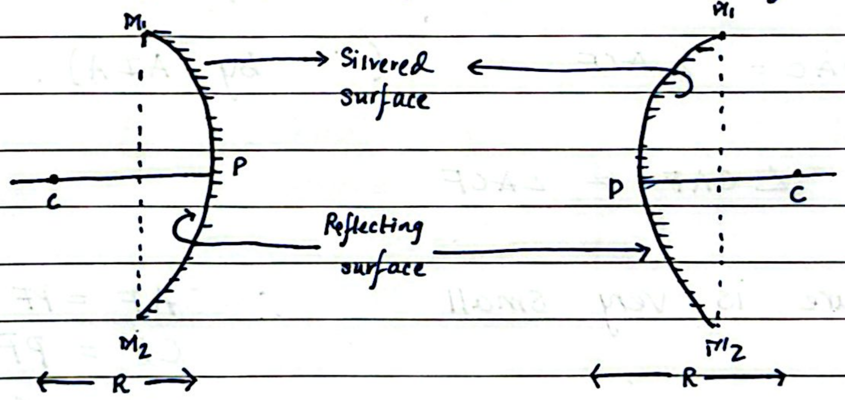
Chapter ~ 9

Ray Optics & Optical Instruments.

⇒ Reflection of light by Spherical Surfaces



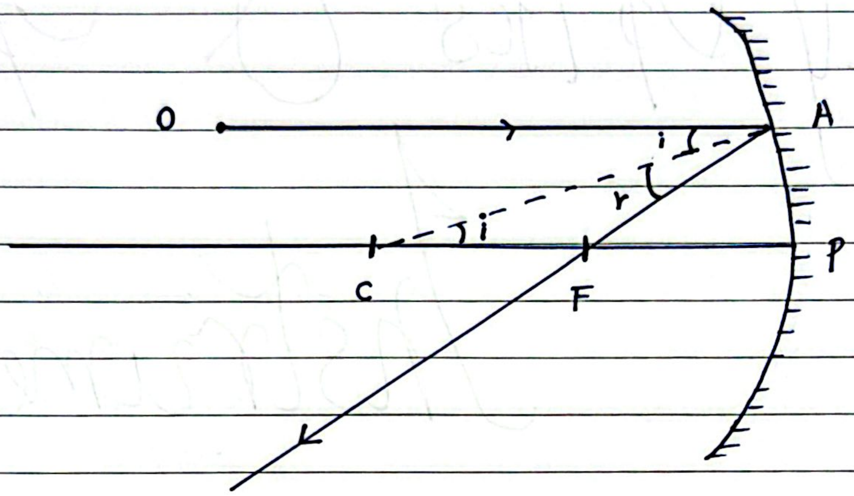
Concave Mirror   Convex mirror

Sign convention : All distances measured from pole

- Distances measured in direction of incident light (rightword if it travels left-right) are positive against it are negative.
- Heights above principal axis are positive below principal axis are negative

⟹ Relation between focal length (f) and Radius of Curvature (R)

To Prove : $f = R/2$



$\angle OAC = \angle CAF$ ($\because$ by $i=r$ law).

$\angle OAC = \angle ACF$ ($\because$ by AIA).

$\therefore \angle CAF = \angle ACF$

Aperture is very small $\therefore$ $AF = PF$

$CF = PF$

$\therefore CF = f$

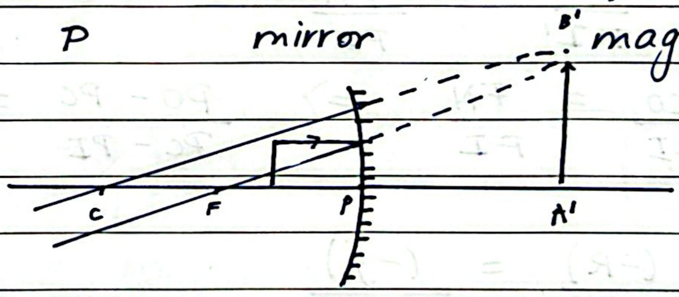
$PC = f + f$

$R = 2f$

$\boxed{\frac{R}{2} = f}$

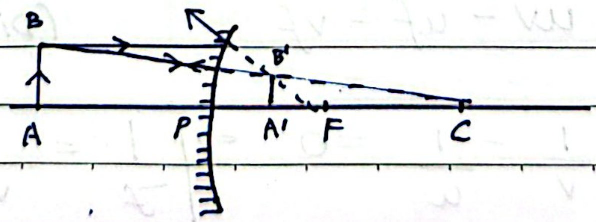
=> Position and Nature of image formed by Spherical Mirrors.

Position of object	Position of Image	Nature
(A) Concave Mirror		
=> At Infinity	At Focus	Real, Inverted Highly diminished.
=> Beyond C	Between C and F	Real, Inverted Diminished.
=> At C	At C	Real, Inverted same size.
=> Between C and F	Between C and ∞	Real, Inverted magnified.
=> At F	At Infinity	Real, Inverted highly magnified
=> Between F and P	Behind the mirror	Virtual Erect magnified.

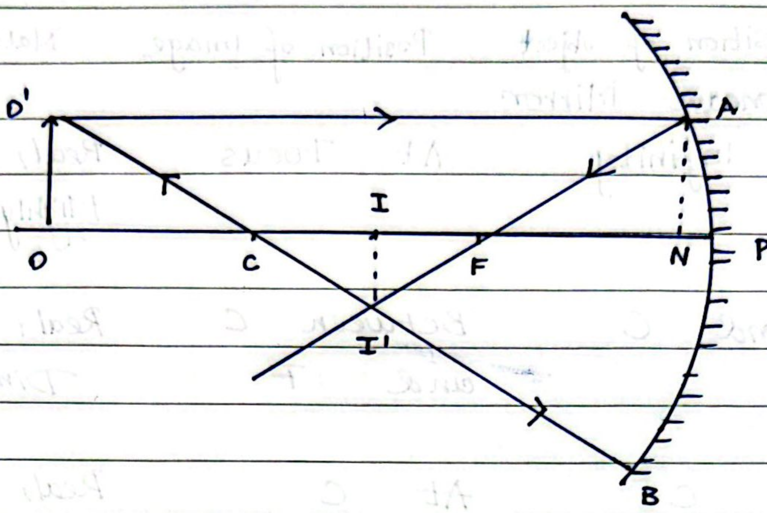


(B) Convex Mirror

Position of object	Position of Image	Nature
=> At Infinity	At F	Virtual, erect highly diminished
=> At finite distance	Between P and F	Virtual, erect, diminished.



⇒ Mirror formula for Concave mirror



Take $\Delta OO'C$ & $\Delta I'I'C$	Take $\Delta II'F$ & ΔNAF
$\angle C = \angle C$ (VOA)	$\angle F = \angle F$ (VOA)
$\angle O = \angle I$ (90°)	$\angle I = \angle N$ (90°)
$\therefore \Delta OO'C \sim \Delta II'C$ (By AA)	$\therefore \Delta II'F \sim \Delta NAF$ (By AA)
$\frac{OO'}{II'} = \frac{O'C}{I'C} = \frac{CO}{CI}$	$\frac{II'}{NA} = \frac{I'F}{AF} = \frac{FI}{FN}$

$$\therefore \frac{OO'}{II'} = \frac{FN}{FI} \quad (NF = PF)$$

Now, $\frac{CO}{CI} = \frac{FN}{FI} \Rightarrow \frac{PO - PC}{PC - PI} = \frac{PF}{PI - PF}$

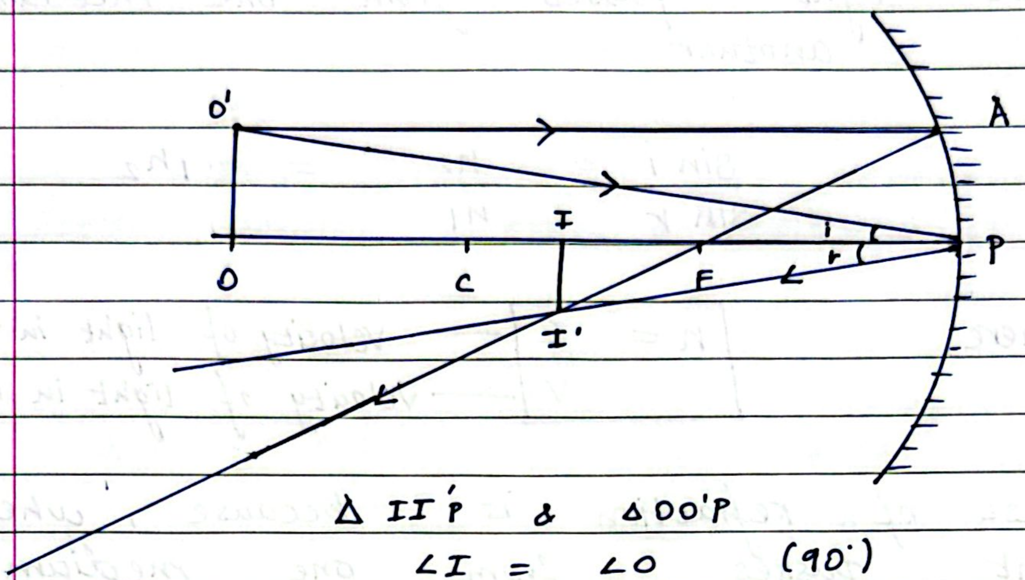
$$\frac{(-u) - (-R)}{(-R) - (-v)} = \frac{(-f)}{(-v) - (-f)}$$

$$\Rightarrow \frac{-u + 2f}{-2f + v} = \frac{-f}{-v + f} \Rightarrow uv - uf - 2fv + 2f^2 = 2f^2 - vf$$

$$uv - uf - vf \quad \text{(Divide } uvf\text{)}$$

$$= \frac{1}{f} - \frac{1}{v} - \frac{1}{u} = 0 \Rightarrow \boxed{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}}$$

⇒ Magnification by Spherical Mirrors.



$\Delta II'P$ & $\Delta OO'P$

$\angle I = \angle O$ (90°)

$\angle P = \angle P$ ($i = r$ By law).

$\Delta II'P \sim \Delta OO'P$ (By AA)

$$\frac{II'}{OO'} = \frac{I'P}{O'P} = \frac{PI}{PO}$$

$$\frac{-I}{+O} = \frac{-v}{-u}$$

$$\boxed{\frac{I}{O} = \frac{-v}{u}}$$

$$\left(\frac{1}{v} + \frac{1}{u} = \frac{1}{f}\right) \times v$$

$$1 - \frac{v}{f} = m$$

$$\boxed{\frac{f-v}{f} = m}$$

$$\left(\frac{1}{v} + \frac{1}{u} = \frac{1}{f}\right) \times u$$

$$1 - \frac{u}{f} = \frac{1}{m}$$

$$\boxed{\frac{f}{f-u} = m}$$

$$\therefore \boxed{m = \frac{-v}{u} = \frac{f-v}{f} = \frac{f}{f-u}}$$

⇒ Reflection of light at a plane Interface.

When light passes from one medium to another

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} = {}_1n_2$$

where, $\boxed{n = \frac{c}{v}}$ c — velocity of light in vacuum, v — velocity of light in medium.

Cause of refraction is because, when light passes from one medium to another its speed is changed.

⇒ Reversibility of light

When a light ray after suffering any number of reflections or refractions has its final path reversed, it travels back along its initial path.

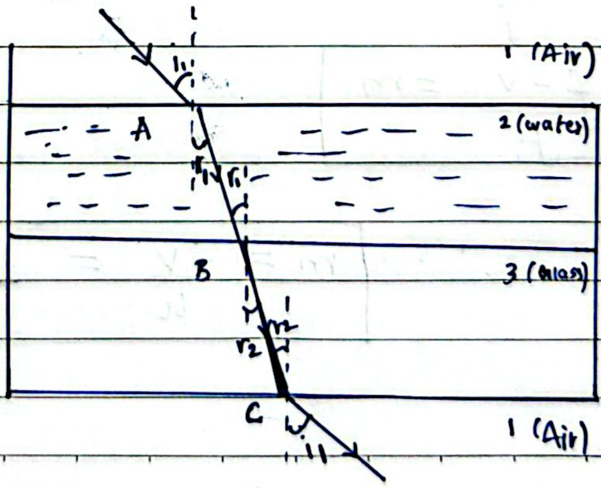
$$\boxed{{}_1n_2 = \frac{1}{{}_2n_1}}$$

⇒ Refraction through parallel multiple media.

$${}_1n_2 \times {}_2n_3 = \frac{1}{{}_3n_1}$$

$$\boxed{\therefore\ {}_1n_3 = {}_1n_2 \times {}_2n_3}$$

$$\boxed{{}_2n_3 = \frac{{}_1n_3}{{}_1n_2}}$$



⇒ Real Depth and Apparent Depth

When an object placed in a denser medium is viewed from rarer medium it appears closer to the surface than it actually is because refracted rays bend away from normal on entering rarer medium.

$$ {}^{a}n_{w} = \frac{R \cdot D}{A \cdot D} $$

$$ A \cdot D = \frac{R \cdot D}{{}^{a}n_{w}} \Rightarrow \boxed{d = R \cdot D\left[1 - \frac{1}{{}^{a}n_{w}}\right]} $$

Apparent shift

$$ \boxed{d = R \cdot D - A \cdot D} $$

⇒ Total Internal Reflection & Critical Angle.

It occurs when light

i] It travels from denser to a rarer medium

ii] Angle of incidence greater than critical angle.

Relation between Refractive Index & Critical Angle

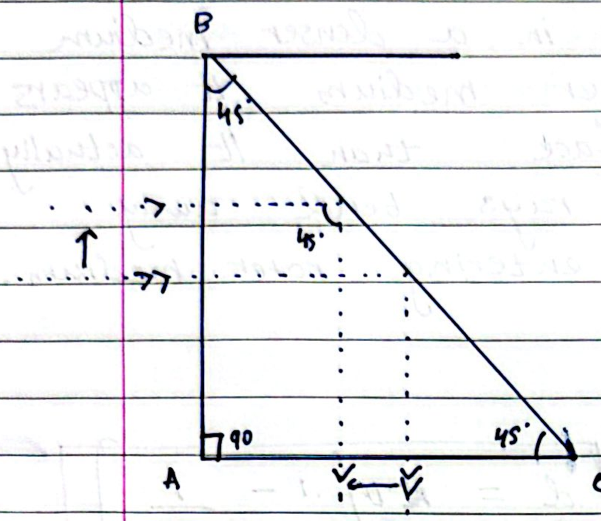
$$ n \sin C = 1 \times \sin 90^\circ $$

$$ n \sin C = 1 $$

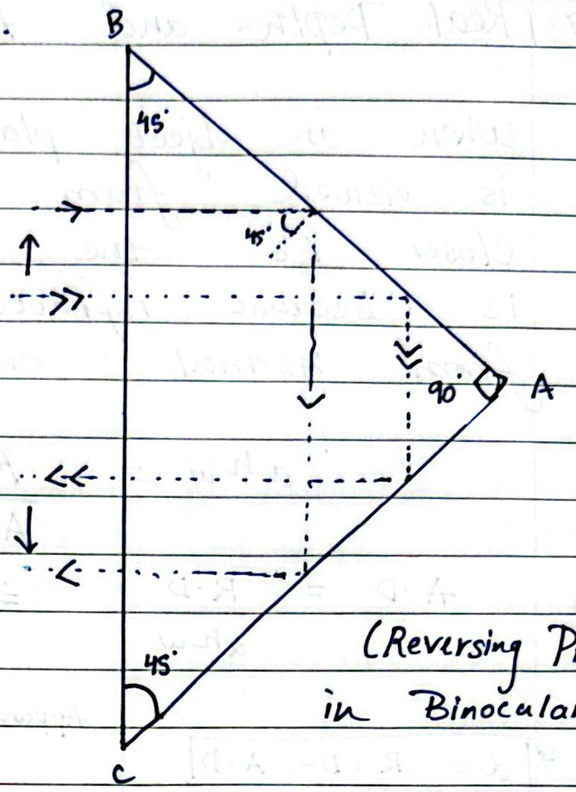
$$ \boxed{\sin C = \frac{1}{n}} $$

⇒ **Optical Fibres**: A thin flexible fibre made of a core of high refractive index glass surrounded by cladding of lower refractive index. It faces multiple TIR along length of fibre over long distances with minimal loss.

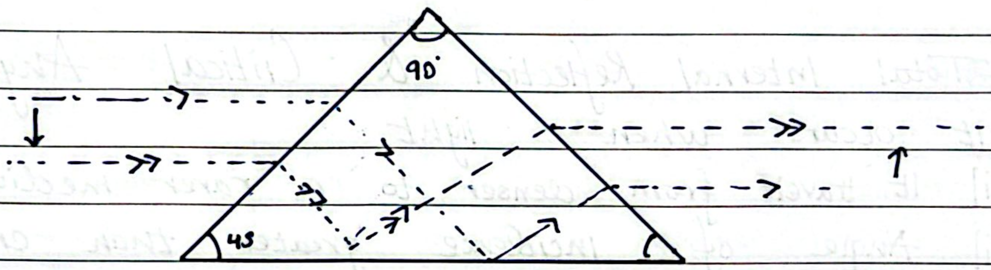
⇒ Total Reflecting Prisms.



(Right Angle Prism)
in periscopes



(Reversing Prism)
in Binoculars

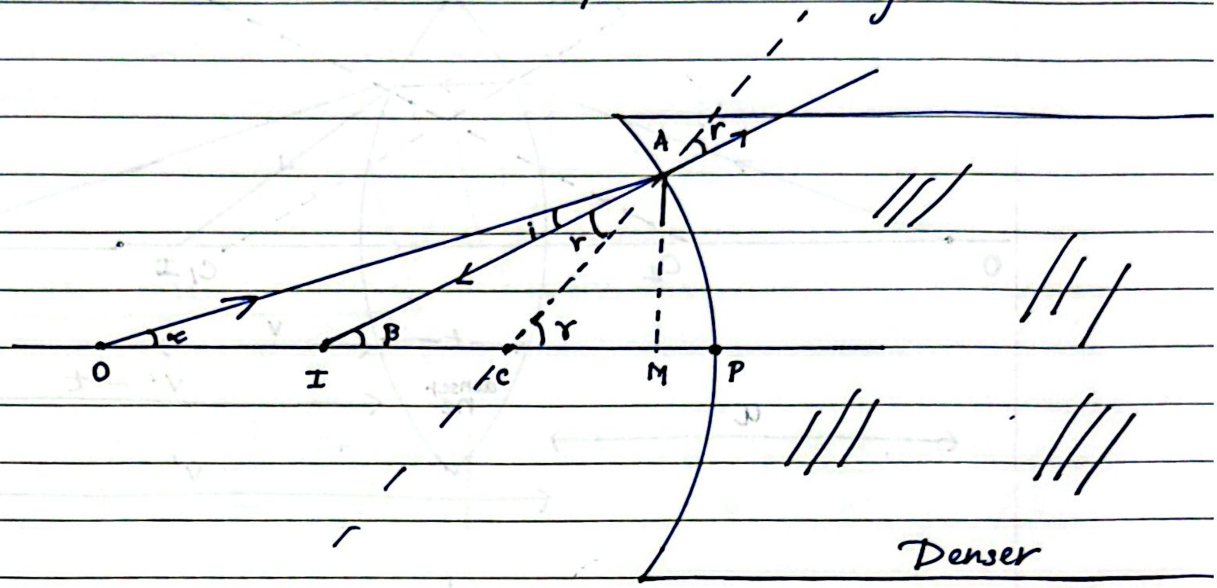


(Erecting Prism)
Prism binoculars.

⇒ Sign convention for u, v and f

⇒ ~~Reflection through a Prism, Minimum deviation.~~
Refraction of light at Spherical surface.

⇒ Refraction at concave spherical surface.



$\alpha + i = \gamma$, $B + r = \gamma$

$i = \gamma - \alpha$, $r = \gamma - B$

$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} \Rightarrow \frac{i}{r} = \frac{n_2}{n_1}$

$\tan\alpha = \frac{AM}{OM} \Rightarrow \alpha = \frac{h}{-u}$

$\tan B = \frac{AM}{IM} \Rightarrow B = \frac{h}{-v}$

$\tan\gamma = \frac{AM}{CM} \Rightarrow \gamma = \frac{h}{-R}$

$$(\gamma - \alpha) n_1 = n_2 (\gamma - B)$$

$$\left(\frac{h}{-R} - \frac{h}{-u}\right) n_1 = n_2 \left(\frac{h}{-R} - \frac{h}{-v}\right)$$

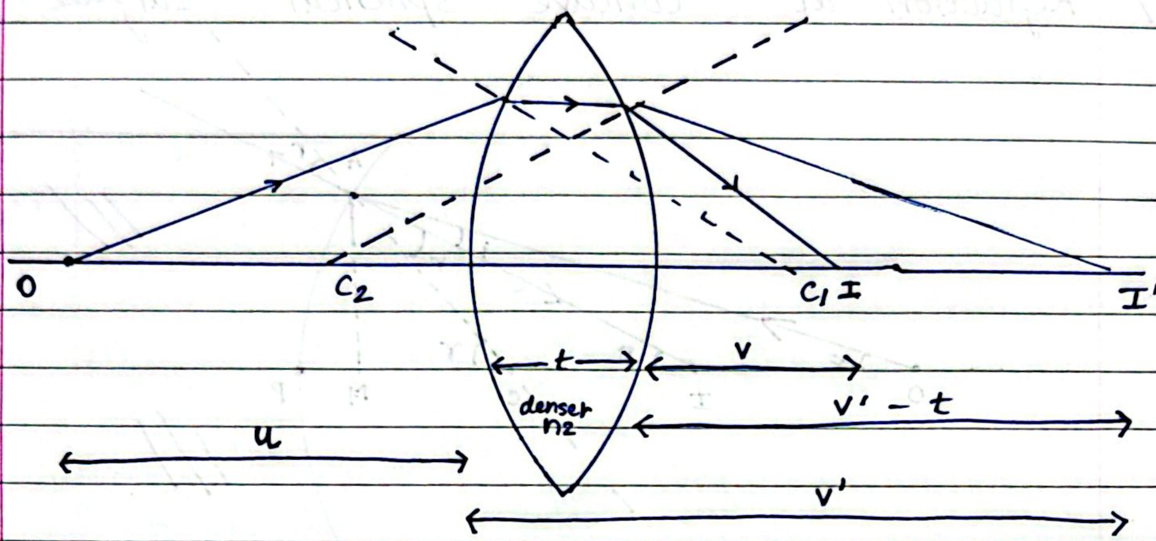
$$\Rightarrow -\frac{n_1}{R} + \frac{n_1}{u} = -\frac{n_2}{R} + \frac{n_2}{v}$$

$$\frac{n_2 - n_1}{R} = \frac{n_2}{v} - \frac{n_1}{u}$$

$$\Rightarrow \boxed{\frac{n-1}{R} = \frac{n}{v} - \frac{1}{u}}$$

$$\boxed{\frac{n_2}{n_1} = n}$$

⟹ Refraction of light through a thin lens (Lens maker Formula)



For Surface 1

$$\frac{n_2-n_1}{R_1} = \frac{n_2}{v'} - \frac{n_1}{u}$$

For Surface 2

$$\frac{n_1-n_2}{R_2} = \frac{n_1}{v} - \frac{n_2}{v'-t}$$

t is very less.

Add Both.

$$\frac{n_2-n_1}{R_1} + \frac{n_1-n_2}{R_2} = \frac{n_1}{v} - \frac{n_2}{u}$$

$$(n_2-n_1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right] = \frac{n_1}{v} - \frac{n_1}{u}$$

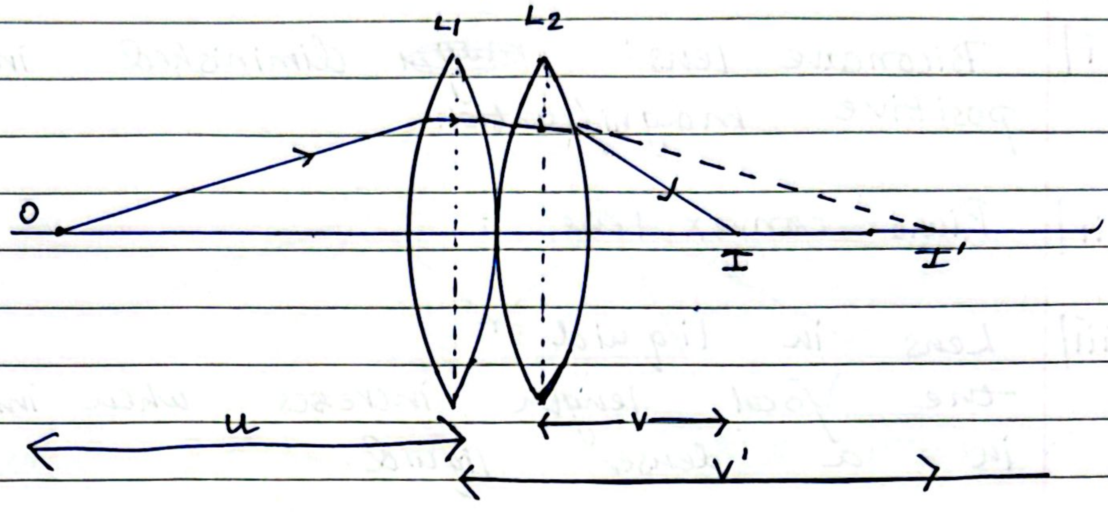
$$\boxed{(n-1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right] = \frac{1}{v} - \frac{1}{u}}$$

⟹ Special Case : if $u=\infty$ then $v=f$

$$(n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{1}{f}$$

⟹ Thin lens formula ⟹ $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

⇒ Combined Focal length of two thin lens

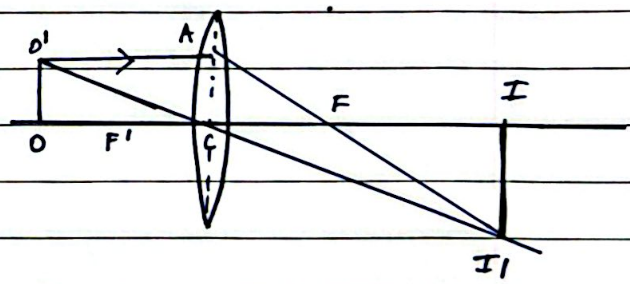


$L_1 = \dfrac{1}{v'} - \dfrac{1}{u} = \dfrac{1}{f_1}$ | $L_2 = \dfrac{1}{v} - \dfrac{1}{v'} = \dfrac{1}{f_2}$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} \quad \text{or} \quad P = P_1 + P_2.$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \boxed{\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}}$$

⇒ Magnification by a biconvex lens



$\Delta CII'$ and $\Delta COO'$ are similar.

$$\frac{II'}{OO'} = \frac{CI}{CO}$$

$$\frac{-h_2}{h_1} = \frac{v}{-u} \Rightarrow \boxed{\frac{h_2}{h_1} = \frac{v}{u} = m}$$

⇒ Types of lens.

i] Biconcave lens: ~~magn~~ diminished image. positive magnification

ii] Plano convex lens. :

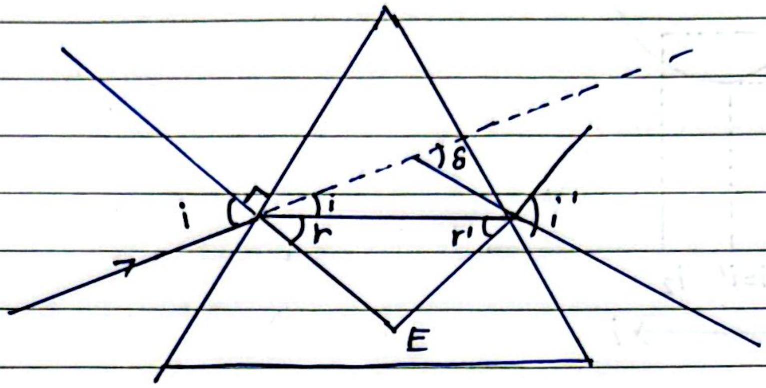
iii] Lens in liquid :
the focal length increases when immersed in a denser liquid.

⇒ Power of a lens.

$$P = \frac{1}{f}$$ unit = Dioptre

convex = +ve power
concave = -ve power.

⇒ Refraction through a Prism.



When $\delta = \delta_m \Rightarrow i' = i$ & $r' = r$

$A + E + 90 + 90 = 360$

$A + E = 180$, $r + r' + E = 180$

$A = 180 - E$ $\quad r + r' = 180 - E$

$\therefore \boxed{A = r + r'}$ $\quad A = 2r$

Now, $(i - r) + (i' - r') = \delta$

$(i + i') - (r + r') = \delta$

$2i - 2r = \delta_{min}$

$\boxed{2i - A = \delta_{min}}$

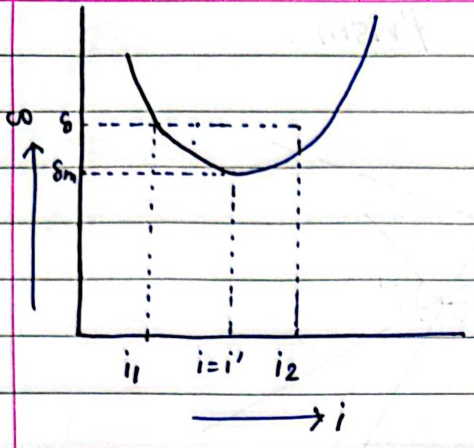
Now, $n = \dfrac{\sin i}{\sin r}$

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin(A/2)}$$

Special Case: thin prism $n = \dfrac{A + \delta_m}{2}$

$\Rightarrow nA = A + \delta_m$

$A(n-1) = \delta_m$



⇒ Dispersion : Splitting of white light into its seven constituent colours.

⇒ Angular Dispersion : Difference between two extreme colours

$$\delta_v - \delta_r = (n_v - n_r)A$$

⇒ Dispersive power (ω)
How strongly materials spreads outs its colours

$$\boxed{\omega = \frac{n_v - n_r}{n_y - 1}}$$

⇒ Rainbow.
Its formed due to dispersion, refraction, total internal reflection.

	Primary Rainbow	Secondary Rainbow.
1.	one TIR and two refractions	two TIR and two refractions.
2.	It is brighter	It is fainter.